WORKED SOLUTION

STEP III 2003, Question 8

Pure • Differential equation

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q8

The key idea

Part (i) is implicit differentiation, but it is really a worked example of a trick: a family of curves $(y+ax)^n(y+bx)=c$ has a gradient that is a ratio of two linear expressions in x and y. Part (ii) runs the trick backwards. We differentiate the general family, read off the gradient in terms of a, b and n, and choose them to match the given equation. Once they match, the expression is constant along every solution curve, and that constant relationship is the general solution.

Solution

Part (i)

Differentiate $(y+2x)^3(y-4x)=c$ with respect to x, using the product rule and the chain rule, and write y' for $\frac{\mathrm{d}y}{\mathrm{d}x}$:

$$3(y+2x)^2(y'+2)(y-4x)+(y+2x)^3(y'-4)=0.$$

At points with $y+2x\neq 0$ we can divide by $(y+2x)^2$:

$$3(y'+2)(y-4x)+(y+2x)(y'-4)=0.$$

Collecting the terms in y',

$$\begin{aligned} y'[3(y-4x)+(y+2x)] &= -6(y-4x)+4(y+2x), \\ y'(4y-10x) &= 32x-2y, \end{aligned}$$

and dividing top and bottom by 2,

$$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{16x-y}{2y-5x},$$

as required. (On the line $y=-2x$ the curve has $c=0$, and there the formula still gives the right gradient, -2.)

Part (ii)

Let $F = (y+ax)^n(y+bx)$, where y is a function of x. Exactly as in part (i),

$$\begin{aligned}\frac{\mathrm{d}F}{\mathrm{d}x} &= n(y+ax)^{n-1}(y'+a)(y+bx) + (y+ax)^n(y'+b)\\ &= (y+ax)^{n-1}\Big[y'((n+1)y+(nb+a)x) + (na+b)y + (n+1)ab\,x\Big].\end{aligned}$$

So F is constant along a curve precisely when, away from $y=-ax$,

$$y' = -\frac{(na+b)y+(n+1)ab\,x}{(n+1)y+(nb+a)x}. \qquad (*)$$

(Part (i) is the case $a=2$, $b=-4$, $n=3$, which gives $(32x-2y)/(4y-10x)$.)

Ask yourself How do we match $(*)$ with $\dfrac{10x-4y}{3x-y}$, when a fraction can be written in many ways?

A fraction is unchanged when the top and bottom are multiplied by the same number, so we can only match the two up to a common factor. Write the target as $-(4y-10x)/(y-3x)$ and multiply the top and bottom by $k=n+1$, so that the coefficients of y in the denominators agree. We then need

$$na+b=-4(n+1), \qquad (n+1)ab=10(n+1), \qquad nb+a=-3(n+1).$$

The middle equation gives $ab=10$, since $n+1\neq 0$ (if $n=-1$ the denominator of $(*)$ has no y term). Adding the outer two equations gives $(n+1)(a+b)=-7(n+1)$, so $a+b=-7$. Hence a and b are the roots of $t^2+7t+10=0$, namely -2 and -5. Subtracting the outer equations gives

$$(n-1)(a-b)=-(n+1).$$

- With $a=-5$, $b=-2$: $-3(n-1)=-(n+1)$, so $n=2$.
- With $a=-2$, $b=-5$: $3(n-1)=-(n+1)$, so $n=\frac{1}{2}$.

Both choices describe the same curves, since $(y-2x)^{1/2}(y-5x)=c$ squares to $(y-5x)^2(y-2x)=c^2$. We take $n=2$, which avoids square roots.

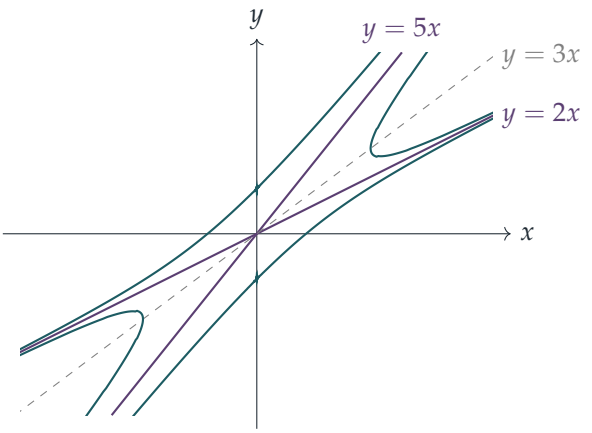
To be sure, check directly. With $G=(y-5x)^2(y-2x)$,

$$\begin{aligned}\frac{\mathrm{d}G}{\mathrm{d}x} &= (y-5x)\big[2(y'-5)(y-2x)+(y-5x)(y'-2)\big]\\ &= (y-5x)\big[3y'(y-3x)-12y+30x\big]\\ &= 3(y-5x)\big[y'(y-3x)-(4y-10x)\big].\end{aligned}$$

If y satisfies the differential equation, then $y'(y-3x)=4y-10x$, so the bracket vanishes and $\mathrm{d}G/\mathrm{d}x=0$. Hence G is constant on every solution curve, and the general solution is

$$\boxed{(y-5x)^2(y-2x)=C}$$

with C an arbitrary constant. The value $C = 0$ gives the straight lines $y = 5x$ and $y = 2x$. Both really are solutions: on $y = 2x$ the right-hand side is $2x/x = 2$, and on $y = 5x$ it is $(-10x)/(-2x) = 5$. The curves all have vertical tangents where they cross the line $y = 3x$, where the equation is not defined.



The curves $(y - 5x)^2(y - 2x) = \pm 1$ (teal) and the line solutions $C = 0$ (plum). The branches between the two lines have a vertical tangent where they meet the dashed line $y = 3x$.

An alternative for part (ii)

The right-hand side depends only on y/x, so the substitution $y = vx$ also works. It gives

$$x\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{(v-2)(v-5)}{3-v},$$

and separating the variables and using partial fractions leads to $\ln|x| = -\frac{1}{3}\ln|v-2| - \frac{2}{3}\ln|v-5| + \text{const}$, which rearranges to the same answer. The method in the question avoids the partial fractions.

Checks

- Part (i) contains a check of $(*)$: putting $a = 2$, $b = -4$, $n = 3$ in it reproduces the gradient found there.
- At the point $(1, 0)$ the equation gives gradient $10/3$. Differentiating $(y - 5x)^2(y - 2x) = C$ implicitly there gives $(-5)\big[2(y' - 5)(-2) + (-5)(y' - 2)\big] = 0$, so $-9y' + 30 = 0$ and $y' = 10/3$.
- Numerically, we stepped the differential equation with a Runge–Kutta method from fifty random starting points. The value of $(y - 5x)^2(y - 2x)$ stayed constant to about 10^{-13} along each path.

Common slips

- Matching the coefficients of $(*)$ with $10x - 4y$ and $3x - y$ directly, without the common factor, which gives inconsistent equations.
- Leaving the answer as $(y - 2x)^{1/2}(y - 5x) = c$ without noticing that it is the same family; it also loses the solutions with $y < 2x$ unless you are careful with signs.
- Forgetting that $C = 0$ is allowed, which gives the line solutions $y = 2x$ and $y = 5x$.

Beyond STEP

The equation here is *homogeneous*: the right-hand side depends only on y/x. Looking for a function $G(x, y)$ that is constant along solutions is the idea behind *exact equations* and *integrating factors*. In mechanics, the same idea is the search for conserved quantities such as energy. Both are covered early in Boyce and DiPrima, *Elementary Differential Equations*.

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